

LLM-Based Football Match Prediction for Sports Betting Markets

Table 1: Core Logic Design of Trading Strategies

Core Logic Design of Trading Strategies		
Targeting Market	Sports betting markets, focusing on football (soccer). Primary market: FIFA World Cup match outcomes (win/draw/loss). Extended markets: major domestic leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1), continental tournaments (Champions League, Copa America, Euros, AFCON), and international qualifiers.	
Phenomena Description ☒	Theoretical Framework ☐	Sports betting odds are set by bookmakers using statistical models but are then adjusted by public betting volume. For high-profile tournaments like the World Cup, casual bettors flood the market with sentiment-driven wagers, pushing odds toward popular narratives (e.g., favoring historically dominant teams, overreacting to recent friendly results). This creates systematic mispricings, particularly for matches involving lesser-known teams with strong underlying player talent, or for group-stage upsets where favorites are overvalued.
Analytical Framework ☒	Theoretical Hypothesis ☐	A fine-tuned LLaMA 3.1 8B model (trained with QLoRA) serves as the quantitative engine. The model ingests structured, player-level statistics from club leagues and international tournaments for both teams in a match. It outputs: (1) predicted match outcome probabilities (win/draw/loss), (2) predicted scoreline, (3) key match events, and (4) a natural-language reasoning chain that references specific player data to justify the prediction. This reasoning chain is what distinguishes the approach from black-box ML: every prediction is auditable, debuggable, and can be overridden by human review before execution.
Phenomena Interpretation ☒	Theoretical Reasoning ☐	The mispricing persists because of three structural factors. First, World Cup and major tournament matches attract disproportionate volumes of recreational bettors whose wagers are driven by national loyalty and media

		narratives rather than data analysis. Second, international football has inherently sparse match history between specific national teams (e.g., Argentina vs. Saudi Arabia had almost no recent head-to-head data before 2022), so bookmaker models rely more on priors and reputation. Third, squad composition changes significantly between tournaments due to retirements, injuries, and emerging talent, meaning historical team-level statistics decay rapidly in predictive value. A model that reasons from current player-level form across domestic leagues can capture these dynamics better than aggregate team-level models.
Trading Opportunity	The trading opportunity arises when the model's predicted probability for a match outcome diverges from the bookmaker's implied probability by more than a minimum edge threshold. For example, if a bookmaker prices Team A to win at implied probability 55%, but the model estimates 72% based on player-level analysis, the 17-point gap represents a value bet. The model also identifies value on underdogs and draws, which are systematically underpriced in public-heavy markets. Backtesting is performed against 2022 FIFA World Cup results (64 matches) as primary ground truth, with extension to 2022 league seasons for robustness validation.	
Trading Operations	Pre-match execution only (no in-play betting). For each match: (1) Assemble structured prompt with both teams' full rosters and per-player statistics. (2) Run inference through the fine-tuned model to generate predictions with reasoning. (3) Human reviews the reasoning chain for factual errors (injured players cited as available, outdated stats, illogical reasoning). (4) Compare model probability against bookmaker implied probability. (5) If edge exceeds confidence threshold, calculate position size via fractional Kelly criterion. (6) Place bet before kickoff. Post-match: record outcome, evaluate reasoning quality, log for future model retraining. Drawdown circuit breaker halts all activity if cumulative losses exceed a predefined bankroll threshold.	
Financial Instrument(s)	Fixed-odds sports betting contracts (1X2: home win / draw / away win) on regulated sportsbooks. Secondary instruments include Asian handicap lines (for matches with a clear favorite, reducing draw variance) and over/under total goals markets (leveraging the model's scoreline predictions).	

Table 2: Design of Algorithmic Models

Design of Algorithmic Models	
Data Requirement	Per-player statistics from API-Football (2018-2024) across major domestic leagues and international tournaments. Features include: goals per 90, assists per 90, pass completion %, key passes, tackles, interceptions, dribble success rate, aerial duels won, minutes played, and overall match rating. Team-level data: squad rosters, formation history, head-to-head records. Ground truth: historical match results with final scores and key events. All data stored as structured CSV files.
Quantitative Model	Fine-tuned LLaMA 3.1 8B using QLoRA (4-bit quantization with Low-Rank Adaptation). The model is instruction-tuned: each training sample is a structured prompt containing both teams' full rosters with per-player statistics, and the target output is the match result, predicted scoreline, key match events, and a reasoning chain explaining the prediction. The model learns to map player-level data patterns to match outcomes while simultaneously learning to articulate its reasoning in natural language. Served locally via vLLM for low-latency inference.
Strategy Type	Event-driven statistical arbitrage. Each match is a discrete event. The model generates an independent probability estimate from player data, and the strategy exploits divergences between this estimate and market-implied probabilities (or conventional expectations). Combines elements of statistical pattern recognition (ML-based) and behavioral exploitation (narrative bias in betting markets).
Trading Decision Triggers	Threshold-based with human-in-the-loop validation. Primary trigger: model's predicted probability for an outcome exceeds a minimum edge threshold above the market-implied probability or baseline expectation. Secondary filter: human review of the model's reasoning chain confirms logical soundness and factual accuracy (no injured players cited, no outdated statistics). A bet is placed only when both the quantitative threshold and reasoning audit are satisfied.
Expected Return Level (1-5)	3. Individual bets can yield 1.5x to 5x+ returns depending on odds, but edge per bet is thin (estimated 5-15% over fair odds). Compounded over a season of league + tournament matches, moderate returns are expected with disciplined position sizing.
Expected Risk Level (1-5)	4. Sports betting is inherently high variance. Single-match outcomes are noisy, upsets happen, and even a well-

Expected Capacity Level (1-5)	calibrated model will lose many individual bets. Risk is managed through fractional Kelly sizing and bankroll caps, but short-term drawdowns of 20-30% are realistic.
	2. Sports betting markets have limited liquidity compared to equity or forex markets. Large wagers move the line and erode edge. World Cup and top league matches have higher liquidity, but the strategy is best suited for small to medium bankrolls.
Expected Frequency Level (1-5)	3. Scoped to World Cup only, frequency is very low (64 matches every 4 years). However, the model generalizes to major domestic leagues and continental tournaments, yielding hundreds to thousands of matches per year. Premier League alone has 380 matches per season. With 5+ top leagues, Champions League, and international windows, the strategy can generate signals on 2,000+ matches annually.
Parameters* (definition and possible ranges)	Trading Parameters: Edge Threshold (T_{edge}): minimum probability gap to trigger a bet. Range: 5-20%. Higher values reduce trade count but increase average edge per trade. Kelly Fraction (f): fraction of full Kelly criterion used for position sizing. Range: 0.1-0.5. Lower values sacrifice returns for bankroll preservation. Max Exposure Per Match (E_{max}): maximum percentage of bankroll wagered on a single match. Range: 1-5%. Drawdown Circuit Breaker (D_{stop}): cumulative loss threshold that pauses all betting. Range: 10-25% of bankroll. Model Hyperparameters: Base Model: LLaMA 3.1 8B. LoRA Rank (r): 16-64. LoRA Alpha: 16-64. Learning Rate: $1e-4$ to $3e-4$. Training Sequence Length: 768-2048 tokens. Quantization: 4-bit (NF4). Inference Temperature: 0.1-0.7 (lower for more deterministic predictions).